



Random sampling of signals concentrated on compact set in localized reproducing kernel subspace of $L^p(\mathbb{R}^n)$

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Received: 8 September 2021 / Accepted: 14 September 2023 / Published online: 17 October 2023
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Abstract

The paper is devoted to studying the stability of random sampling in a localized reproducing kernel space. We show that if the sampling set on Ω (compact) discretizes the integral norm of simple functions up to a given error, then the sampling set is stable for the set of functions concentrated on Ω . Moreover, we prove with an overwhelming probability that $\mathcal{O}(\mu(\Omega)(\log \mu(\Omega))^3)$ many random points uniformly distributed over Ω yield a stable set of sampling for functions concentrated on Ω .

Keywords Random sampling · Reproducing kernel space · Sampling inequality · Covering number · Discretization

Mathematics Subject Classification (2010) Primary 42A61 · 94A20 · 41A65, Secondary 26D15 · 60E15

1 Introduction

The sampling problem is one of the most active research area in the field of signal processing, image processing, and digital communication. The problem is related to find a discrete sample set such that a function f can be uniquely determined and reconstructed by its discrete sample values. However, this problem is not well defined unless we assume some additional information on the function space. In this paper, we focus on the space of localized reproducing kernel subspace of $L^p(\mathbb{R}^n) := L^p(\mathbb{R}^n, \mu)$, where μ is a Lebesgue measure on $\mathbb{R}^n$.

Communicated by: Felix Krahmer

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A countable set $X = \{x_j \in \mathbb{R}^n : j \in J\}$ is said to be a stable sample or stable set of sampling for the function space $V \subseteq L^p(\mathbb{R}^n)$ if there exist positive constants A and B such that

$$A \|f\|_{L^p(\mathbb{R}^n)}^p \leq \sum_{j \in J} |f(x_j)|^p \leq B \|f\|_{L^p(\mathbb{R}^n)}^p \quad \forall f \in V. \quad (1.1)$$

In the case of Paley-Wiener space $PW_{[a,b]}(\mathbb{R})$, Beurling density condition characterizes the stability of a sampling set. However, a similar result is not valid for $PW_S(\mathbb{R}^n)$, where S is a convex subset of $\mathbb{R}^n$, see [31, Section 5.7]. Hence, to overcome these difficulties in studying non-uniform sampling in higher-dimension, we consider a set of random points and check the probability of a stable sampling set. At the same time, the problem of finding a stable random sampling on $\mathbb{R}^n$ is not feasible in general. Bass and Gröchenig [3] observed that for each random sample identically and uniformly distributed over each cube $k + [0, 1]^n$ in $\mathbb{R}^n$, the sampling inequality (1.1) fails almost surely for Paley-Wiener space. Moreover, for smooth function f in $L^p(\mathbb{R}^n)$, the sample value $f(x_j)$ may not assist in sampling inequality for large values of x_j . To resolve these problems, we consider random sample points are drawn uniformly and identically from a compact set Ω , and a class of functions concentrated on Ω . Such functions are useful in many applications of engineering fields such as information and communication theory [13], signal detection and estimation [21], neuroscience [22], and optics [17].

Random sampling problem is closely related to learning theory [9, 34, 36] and compressed sensing [15] and widely applied in information recovery [25]. Trigonometric polynomials are effectively used in practical applications such as computer tomography [1], geophysics [35], image processing [37], and cardiology [38]. Bass and Gröchenig studied random sampling for multivariate trigonometric polynomial [2]; Candés, Romberg, and Tao reconstructed sparse trigonometric polynomial from a random sample set [7]. In the last decades, random sampling was studied for Paley-Wiener space [3, 4]; shift-invariant space [18, 40, 42]; continuous function space with bounded derivative [41]; function space with finite rate of innovation [29]; and reproducing kernel subspace of $L^p(\mathbb{R}^n)$ which is an image of an idempotent integral operator [28, 32]. The related random sampling problem was studied in weighted trigonometric polynomials [27], and spherical harmonic polynomials [33]. The worst-case error analysis of sampling recovery of function from their random sample values has been addressed in [14, 23, 39].

In this paper, we consider localized reproducing kernel subspace V of $L^p(\mathbb{R}^n)$ (defined in Sect. 2), which takes into consideration of existing model spaces. A subset $V^*(\Omega, \delta)$ of V is a set of δ -concentrated functions, defined as follows

$$V^*(\Omega, \delta) := \left\{ f \in V : \int_{\Omega} |f(x)|^p dx \geq (1 - \delta) \|f\|_{L^p(\mathbb{R}^n)}^p \right\}, \quad 0 < \delta < 1.$$

We are interested in finding probability bound for random sample $\{\xi_\nu : \nu = 1, 2, \dots, r\}$ uniformly and identically drawn from Ω to be a stable sample set for

$V^*(\Omega, \delta)$ and satisfy the sampling inequality

$$A\|f\|_{L^p(\mathbb{R}^n)}^p \leq \frac{1}{r} \sum_{v=1}^r |f(\xi_v)|^p \leq B\|f\|_{L^p(\mathbb{R}^n)}^p \quad \forall f \in V^*(\Omega, \delta). \quad (1.2)$$

Of course, the sampling inequality can be achievable with high probability for large-scale sample size. However, the problem seems interesting if one can find a minimal sample size required to satisfy the sampling inequality (1.2). It was proved for δ -concentrated functions on the cube $C_R := [-R/2, R/2]^n$ that sample size required to be of $\mathcal{O}(R^{2n})$ in the case of Paley-Wiener space [3], shift-invariant space [40], and image of an idempotent integral operator [32]. However, in recent years, it was shown that effective number of sample is indeed of order $\mathcal{O}(R^n \log R^n)$ for Paley-Wiener space [4] and shift-invariant subspace of $L^2(\mathbb{R}^n)$ [18]. The recent article [28] by Li et al. proved that for the space of image of an idempotent integral operator, δ -concentrated functions on Corkscrew domain Ω satisfy (1.2) with sample size of order $\mathcal{O}(\nu(\Omega) \log \nu(\Omega))$, where ν denotes the metric measure on Ω .

The main features of this paper are summarized as follows:

- (i) In the recent articles [28, 32], the random sampling problem studied for an image of an idempotent integral operator T , and the stability of the sampling set depends on the regularity of the kernel, i.e.,

$$\lim_{\epsilon \rightarrow 0} \|w_\epsilon(K)\|_S = 0, \quad (1.3)$$

where $w_\epsilon(K)(x, y) = \sup_{x', y' \in [-\epsilon, \epsilon]^n} |K(x + x', y + y') - K(x, y)|$, and

$$\|K\|_S := \max \left(\sup_{x \in \mathbb{R}^n} \|K(x, \cdot)\|_{L^1(\mathbb{R}^n)}, \sup_{y \in \mathbb{R}^n} \|K(\cdot, y)\|_{L^1(\mathbb{R}^n)} \right).$$

For example, the quasi-shift invariant space can be viewed as the image of an idempotent integral operator with the kernel having product quasi-shifts and their corresponding biorthogonal dual frame. In literature, there is no explicit construction of dual frame for frame having quasi-shift. As a consequence, the regularity condition (1.3) cannot be verified, see Example 2.2 for more details. In this paper, we drop this assumption and study the random sampling problem for the localized reproducing kernel space, which generalizes the image space of an idempotent integral operator, quasi-shift invariant space, and weighted Fock space.

- (ii) We show that any element in V can be approximated by an element in a finite-dimensional subspace of V . As a consequence, we can prove the sampling inequality (1.2) with high probability using the same line of proof in [3, 32, 42] if the sample size is of order $\mathcal{O}(\mu(\Omega)^2)$, where Ω is a bounded cube in $\mathbb{R}^n$. The probability bound of the sampling inequality (1.2) can be achieved with the sample size of $\mathcal{O}(\mu(\Omega) \log \mu(\Omega))$ by estimating the decay of eigenvalues of the self-adjoint compact operator [4, 18, 29] or by the sampling density condition

using the kernel regularity assumption [28]. In this paper, by weakening the regularity assumption of the reproducing kernel and considering compact subset Ω of $\mathbb{R}^n$, we solve the random sampling problem for the localized reproducing kernel subspace of $L^p(\mathbb{R}^n)$ using the idea of Marcinkiewicz type discretization result introduced in [11]. We show that if a sampling set is “good” discretization to the integral norm on Ω for the class of simple functions, then it is a stable sampling set for δ -concentrated functions on Ω . Furthermore, we prove that the sampling inequality (1.2) holds with high probability if the sample size is of order $\mathcal{O}(\mu(\Omega)(\log \mu(\Omega))^3)$.

We pursue the approach of [11, 40] with a mild condition on generators. Note that the result in [40] based on strong decay condition of generators

$$\phi(x) \leq \frac{C}{(1 + |x|)^m}, \quad x \in \mathbb{R}^n,$$

and in [11] relied on the assumption of boundedness of entropy number.

This paper is organized as follows. In Sect. 2, we give basic definitions, notations, and preliminary results. In Sect. 3, we show that functions in a given compact set are bounded by simple functions. Moreover, under some condition on simple functions, we show that the sampling inequality holds for functions concentrated in a compact set. The main result of this paper is provided in Sect. 4.

2 Preliminaries

In this section, we define the localized reproducing kernel subspace V of $L^p(\mathbb{R}^n)$, and discuss some interesting examples.

A sample set $U = \{u : u \in \mathbb{R}^n\}$ is *relatively separated* with gap β if

$$\beta := \inf_{\substack{u, u' \in U \\ u \neq u'}} \|u - u'\|_\infty > 0,$$

and there exists $\epsilon > 0$ such that $\cup_{u \in U} B(u; \epsilon) = \mathbb{R}^n$, where $B(x; r)$ denotes an open ball of radius r centered at x .

The *Wiener amalgam space* $W(L^1)(\mathbb{R}^n)$ consists of all functions $f \in L^\infty(\mathbb{R}^n)$ such that

$$\|f\|_{W(L^1)(\mathbb{R}^n)} := \sum_{k \in \mathbb{Z}^n} \sup_{x \in [0, 1]^n} |f(x + k)| < \infty.$$

A family $\{x_i : i \in I\}$ of elements of a Hilbert space H is called a *frame* for H if there exist constants $c_1, c_2 > 0$ such that

$$c_1 \|x\|^2 \leq \sum_{i \in I} |\langle x, x_i \rangle|^2 \leq c_2 \|x\|^2 \quad \forall x \in H.$$

Definition 2.1 We say that a closed subspace $W \subseteq L^2(\mathbb{R}^n)$ is *localizable reproducing kernel Hilbert space*, if there exist:

- (a) A relatively separated set $\Gamma \subseteq \mathbb{R}^n$ (nodes);
- (b) A function $\Theta \in W(L^1)(\mathbb{R}^n)$ (envelope);
- (c) A collection of continuous functions $\{F_\gamma : \gamma \in \Gamma\}$ is a frame for W , and satisfy the localization estimate

$$|F_\gamma(x)| \leq \Theta(x - \gamma), \quad \forall \gamma \in \Gamma. \quad (2.1)$$

The coefficient map $f \mapsto Cf := (\langle f, F_\gamma \rangle)_{\gamma \in \Gamma}$ is bounded from $L^2(\mathbb{R}^n) \rightarrow \ell^2(\Gamma)$, and can be extended to the bounded operator $C : L^p(\mathbb{R}^n) \rightarrow \ell^p(\Gamma)$. Likewise, the adjoint operator $C^* : \ell^p(\Gamma) \rightarrow L^p(\mathbb{R}^n)$, $1 \leq p < \infty$ defined by $c \mapsto C^*c := \sum_{\gamma \in \Gamma} c_\gamma F_\gamma$ is bounded. Using the collection $\{F_\gamma : \gamma \in \Gamma\}$ as a frame, we get the range space

$$V := C^*(\ell^p(\Gamma)) = \left\{ \sum_{\gamma \in \Gamma} c_\gamma F_\gamma : c \in \ell^p(\Gamma) \right\}$$

is a well-defined closed subspace of $L^p(\mathbb{R}^n)$, for $1 \leq p < \infty$, and there exist constants $A, B > 0$ such that

$$A \|f\|_{L^p(\mathbb{R}^n)}^p \leq \sum_{\gamma \in \Gamma} |c_\gamma|^p \leq B \|f\|_{L^p(\mathbb{R}^n)}^p, \quad (2.2)$$

for every $f = \sum_{\gamma \in \Gamma} c_\gamma F_\gamma \in V$ (see [19]). Moreover, it is easy to show that V is a reproducing kernel Banach space. First, let us recall a basic definition.

Definition 2.2 A Banach space Σ of functions on a set X is a *reproducing kernel Banach space* if the point evaluation functional $f \mapsto f(x)$ is continuous for each $x \in X$, i.e., for every $x \in X$, there exists $C_x > 0$, such that $|f(x)| \leq C_x \|f\|$, for all $f \in \Sigma$.

Lemma 2.3 The space V is a reproducing kernel Banach space.

Proof Let $x \in \mathbb{R}^n$ be fixed and $f \in V$ be arbitrary.

$$\begin{aligned} f(x) &= \sum_{\gamma \in \Gamma} c_\gamma F_\gamma(x), \\ |f(x)| &\leq \left(\sum_{\gamma \in \Gamma} |c_\gamma|^p \right)^{\frac{1}{p}} \left(\sum_{\gamma \in \Gamma} |F_\gamma(x)|^{p'} \right)^{\frac{1}{p'}} \end{aligned}$$

$$\begin{aligned}
&\leq B^{\frac{1}{p}} \|f\|_{L^p(\mathbb{R}^n)} \left(\sum_{\gamma \in \Gamma} |\Theta(x - \gamma)|^{p'} \right)^{\frac{1}{p'}} \\
&\leq B^{\frac{1}{p}} \|f\|_{L^p(\mathbb{R}^n)} \left(\sum_{\gamma \in \Gamma} |\Theta(x - \gamma)| \right)^{\frac{1}{p'}} \\
&= B^{\frac{1}{p}} \|f\|_{L^p(\mathbb{R}^n)} \left(\sum_{k \in \mathbb{Z}^n} \sum_{\gamma \in \Gamma \cap [k, k+1]^n} |\Theta(x - \gamma)| \right)^{\frac{1}{p'}} \\
&\leq B^{\frac{1}{p}} \|f\|_{L^p(\mathbb{R}^n)} \left(\sum_{k \in \mathbb{Z}^n} N(\Gamma) \sup_{x \in [0, 1]^n} |\Theta(x - k)| \right)^{\frac{1}{p'}} \\
|f(x)| &\leq B^{\frac{1}{p}} N(\Gamma)^{\frac{1}{p'}} \|\Theta\|_{W(L^1)(\mathbb{R}^n)}^{\frac{1}{p'}} \|f\|_{L^p(\mathbb{R}^n)},
\end{aligned}$$

where $N(\Gamma) = \sup_{k \in \mathbb{Z}^n} |\Gamma \cap (k + [0, 1]^n)|$ and $|X|$ is the cardinality of the set X . Therefore, the point evaluation functional is continuous for each $x \in \mathbb{R}^n$. $\square$

From the proof of the above lemma, we choose a constant $D > 1$ such that

$$\|f\|_{L^\infty(\mathbb{R}^n)} \leq D \|f\|_{L^p(\mathbb{R}^n)}, \quad \forall f \in V. \quad (2.3)$$

Example 2.1 In the following, we give some well-known examples of localizable reproducing kernel spaces.

1. Let $\phi \in W(L^1)(\mathbb{R}^n)$ be such that for some positive constants A_1 and B_1 ,

$$A_1 \leq \sum_{k \in \mathbb{Z}^n} |\hat{\phi}(\xi + k)|^2 \leq B_1.$$

Then, $\{\phi(\cdot - k) : k \in \mathbb{Z}^n\}$ is a frame for the shift-invariant space $V(\phi) = \left\{ \sum_{k \in \mathbb{Z}^n} c_k \phi(\cdot - k) : (c_k) \in \ell^2(\mathbb{Z}^n) \right\}$, see [8, Theorem 9.2.5]. In this case, we have $F_\gamma = \phi(\cdot - \gamma)$, $\gamma \in \Gamma = \mathbb{Z}^n$.

2. Let T be an idempotent integral operator ($T^2 = T$) on $L^p(\mathbb{R}^n)$ with the integral kernel K satisfies the off-diagonal decay condition $\|K\|_S < \infty$ and the regularity condition (1.3). Then, the image space V of T is a reproducing kernel subspace of $L^p(\mathbb{R}^n)$, and there exist relatively separated set $\Gamma \subseteq \mathbb{R}^n$ and the collection $\{\phi_\gamma : \gamma \in \Gamma\}$ forms p -frame for V . Apart from that, there exists $h \in W(L^1)(\mathbb{R}^n)$ such that $|\phi_\gamma(x)| \leq h(x - \gamma)$, see [30].

So far, the sampling problem had been studied for the image of an idempotent integral operator in [28, 30, 32] and the regularity condition (1.3) was the key assumption. However, in practice, the condition is not feasible. In the following, we give examples of localizable reproducing kernel space, which is the image of an idempotent integral operator but the regularity condition is either hard to verify or does not satisfy.

Example 2.2 Let $X = \{x_k : k \in \mathbb{Z}\}$ be a relatively separated set of strictly increasing sequence in $\mathbb{R}$ and $\phi \in W(L^1)(\mathbb{R})$. A quasi-shift invariant space is defined by

$$V_X(\phi) = \left\{ f = \sum_{k \in \mathbb{Z}} c_k \phi(\cdot - x_k) : c = (c_k) \in \ell^2(\mathbb{Z}) \right\}.$$

For more literature on quasi-shift invariant space, we refer the interested reader to [16, 24]. If the collection $\{\phi(\cdot - x_k) : k \in \mathbb{Z}\}$ is a frame for $V_X(\phi)$, then the space $V_X(\phi)$ is a localizable reproducing kernel Hilbert space. Furthermore, $V_X(\phi)$ can be written as the image of an idempotent integral operator with the integral kernel

$$K(x, y) = \sum_{k \in \mathbb{Z}} \overline{\phi(x - x_k)} \tilde{\phi}_k(y),$$

where $\{\tilde{\phi}_k : k \in \mathbb{Z}\}$ is the canonical dual frame of $\{\phi(\cdot - x_k) : k \in \mathbb{Z}\}$. In the case of irregular sample set X , there is no constructive method to compute $\tilde{\phi}_k$. As a consequence, the condition on K is not easily verifiable. Even for the simplest case, the kernel assumption (1.3) was not satisfied. For example, let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$\phi(x) = \begin{cases} 1, & \text{if } 0 \leq x \leq \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\phi \in W(L^1)(\mathbb{R})$ and $\|\phi\|_{W(L^1)(\mathbb{R})} = 1$. The collection $\{\sqrt{2}\phi(\cdot - k) : k \in \mathbb{Z}\}$ is an orthonormal basis of the shift-invariant space $V(\phi) \subset L^2(\mathbb{R})$ and the reproducing kernel of $V(\phi)$ is defined by $K(x, y) = \sum_{k \in \mathbb{Z}} 2\phi(x - k)\phi(y - k)$.

For $\epsilon < \frac{1}{2}$,

$$\sup_{x' \in [-\epsilon, \epsilon]} |K(x + x', y) - K(x, y)| = \begin{cases} 2\phi(y - k), & \text{if } x \in (k - \epsilon, k + \epsilon) \cup (k + \frac{1}{2} - \epsilon, k + \frac{1}{2} + \epsilon), \\ 0, & \text{otherwise.} \end{cases}$$

Hence,

$$\left\| \sup_{x' \in [-\epsilon, \epsilon]} |K(x + x', \cdot) - K(x, \cdot)| \right\|_{L^1(\mathbb{R})} = \begin{cases} 1, & \text{if } x \in (k - \epsilon, k + \epsilon) \cup (k + \frac{1}{2} - \epsilon, k + \frac{1}{2} + \epsilon), \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\begin{aligned} 1 &= \sup_{x \in \mathbb{R}} \left\| \sup_{x' \in [-\epsilon, \epsilon]} |K(x + x', \cdot) - K(x, \cdot)| \right\|_{L^1(\mathbb{R})} \\ &\leq \max \left(\sup_{x \in \mathbb{R}} \|w_\epsilon(K)(x, \cdot)\|_{L^1(\mathbb{R})}, \sup_{y \in \mathbb{R}} \|w_\epsilon(K)(\cdot, y)\|_{L^1(\mathbb{R})} \right) = \|w_\epsilon(K)\|_S. \end{aligned}$$

We collect our assumption on compact domain Ω and the function Θ in the following list:

Assumption 1 Without loss of generality, assume that $\mu(\Omega) \geq 1$, and denote d as the number of unit cube of the form $m + [0, 1]^n$ covers the boundary of Ω , where $m \in \mathbb{Z}^n$. Let the number of Γ in Ω be bounded by $C(\Gamma)\mu(\Omega)$, where $C(\Gamma)$ is some positive constant.

Assumption 2 For every $C_N = [-N/2, N/2]^n$,

$$\sum_{k \in \mathbb{Z}^n \setminus C_N} \sup_{x \in [0, 1]^n} |\Theta(x - k)| < \frac{C}{N^{n\alpha}}, \quad (2.4)$$

where C is a positive constant, and $\alpha \geq \frac{p}{p-1}$ for $1 < p < \infty$ and $\alpha \geq 1$ for $p = 1$.

In the following, we provide an example of localizable reproducing kernel space satisfying Assumption 2.

Example 2.3 Let $\phi : \mathbb{C}^n \rightarrow \mathbb{R}$ be a plurisubharmonic function and there exist $m, M > 0$ such that

$$im\partial\bar{\partial}|z|^2 \leq i\partial\bar{\partial}\phi \leq iM\partial\bar{\partial}|z|^2, \quad (2.5)$$

in the sense of positive currents [26]. Let A_ϕ^2 be the space of entire function on $\mathbb{C}^n$ equipped with the norm

$$\|f\|_{\phi, 2}^2 := \int_{\mathbb{C}^n} |f(z)|^2 e^{-2\phi(z)} dz.$$

Then, A_ϕ^2 is a reproducing kernel Hilbert space. Let K_ϕ be the reproducing kernel of A_ϕ^2 . For the specialized weight $\phi(z) = |z|^2$, the reproducing kernel K_ϕ is known explicitly and has the property

$$K_\phi(z, w)e^{-|z|^2 - |w|^2} = e^{-\frac{1}{2}|z-w|^2}.$$

For the general class of weight, the explicit form of K_ϕ is not known. However, Dellin [12] proved that if the plurisubharmonic function ϕ satisfies (2.5) then the reproducing kernel K_ϕ satisfies the following off-diagonal decay estimate

$$|K_\phi(z, w)|e^{-\phi(z) - \phi(w)} \leq Ce^{-c|z-w|} \quad \forall z, w \in \mathbb{C}^n,$$

for some $c, C > 0$. The weighted Fock space V_ϕ^2 is defined by $V_\phi^2 = \{f = ge^{-\phi} : g \in A_\phi^2\}$, and is a localizable reproducing kernel Hilbert space in $L^2(\mathbb{R}^{2n})$, see [20, Proposition 7.7]. In particular, if $\beta \in (0, \sqrt{2/n})$, then $\Gamma = \beta\mathbb{Z}^{2n}$ is a relatively separated set and there exists frame $\{F_\gamma : \gamma \in \Gamma\}$ for V_ϕ^2 with frame bounds $A = \frac{1}{4\beta^{2n}}$ and $B = \frac{3}{2}$. In addition, the frame $\{F_\gamma : \gamma \in \Gamma\}$ satisfies the localized estimate (2.1), where $\Theta(z) = C\beta e^{-c|z|}$. It is easy to verify that $\Theta \in W(L^1)(\mathbb{R}^{2n})$ and satisfy Assumption 2.

In the following lemma, we show that for any f in V there exists a function in finite-dimensional subspace of V which is close to f . In order to proceed the lemma, let M be a compact set and we define finite-dimensional subspace V_M of V as

$$V_M = \left\{ \sum_{\gamma \in \Gamma \cap M} c_\gamma F_\gamma : c_\gamma \in \mathbb{R} \right\}.$$

Lemma 2.4 *For a given $\epsilon > 0$ and $f \in V$, there exist compact set M (depending on ϵ and Ω) and $\tilde{f} \in V_M$ such that*

$$\|f - \tilde{f}\|_{L^p(\Omega)} < \epsilon \|f\|_{L^p(\mathbb{R}^n)}.$$

Proof Let $f \in V$. Then, there exists $(c_\gamma) \in \ell^p(\Gamma)$ such that $f = \sum_{\gamma \in \Gamma} c_\gamma F_\gamma$. We consider $\tilde{f} = \sum_{\gamma \in \Gamma \cap M} c_\gamma F_\gamma$, where M is a compact subset of $\mathbb{R}^n$ (chosen later).

Now, for $x \in \Omega$,

$$\begin{aligned} |f(x) - \tilde{f}(x)| &\leq \sum_{\gamma \in \Gamma \setminus M} |c_\gamma| |F_\gamma(x)| \\ &\leq \left(\sum_{\gamma \in \Gamma \setminus M} |c_\gamma|^p \right)^{\frac{1}{p}} \left(\sum_{\gamma \in \Gamma \setminus C_N} |F_\gamma(x)|^{p'} \right)^{\frac{1}{p'}} \\ &\leq B^{\frac{1}{p}} \|f\|_{L^p(\mathbb{R}^n)} \left(\sum_{\gamma \in \Gamma \setminus M} |\Theta(x - \gamma)|^{p'} \right)^{\frac{1}{p'}} \\ &\leq B^{\frac{1}{p}} \left(\sum_{\gamma \in \Gamma \setminus M} |\Theta(x - \gamma)| \right)^{\frac{1}{p'}} \|f\|_{L^p(\mathbb{R}^n)}. \end{aligned}$$

As $\Theta \in W(L^1)(\mathbb{R}^n)$ and satisfies (2.4), then for each $\epsilon > 0$ there exists $N = \left(\frac{CN(\Gamma)B^{\frac{p'}{p}}}{\epsilon^{p'}} \right)^{\frac{1}{n\alpha}} \mu(\Omega)^{\frac{1}{n}}$ such that

$$\sum_{k \in \mathbb{Z}^n \setminus C_N} \sup_{x \in [0, 1]^n} |\Theta(x - k)| \leq \frac{C}{N^{n\alpha}} < \frac{\epsilon^{p'}}{N(\Gamma)\mu(\Omega)^{p'} B^{\frac{p'}{p}}}.$$

For $x \in \partial\Omega$, there exists $m \in \mathbb{Z}^n$ such that $x \in m + [0, 1]^n$. Consider

$$\begin{aligned} \sum_{\gamma \in \Gamma \setminus \{m + C_N\}} |\Theta(x - \gamma)| &\leq N(\Gamma) \sum_{k \in \mathbb{Z}^n \setminus C_N} \sup_{y \in [0, 1]^n} |\Theta(y - k)| \\ &< \frac{\epsilon^{p'}}{\mu(\Omega)^{p'} B^{\frac{p'}{p}}}. \end{aligned}$$

Under the Assumption 1, the same is true for every $x \in \partial\Omega \cap \{m_i + [0, 1]^n\}$, where $m_i \in \mathbb{Z}^n$ and the collection $\{m_i + [0, 1]^n : i = 1, 2, \dots, d\}$ cover $\partial\Omega$. Choose $M = \left(\bigcup_{i=1}^d \{m_i + C_N\}\right) \cup \Omega$. Then for each $x \in \Omega$,

$$\sum_{\gamma \in \Gamma \setminus M} |\Theta(x - \gamma)| < \frac{\epsilon^{p'}}{\mu(\Omega)^{p'} B^{\frac{p'}{p}}},$$

and hence

$$|f(x) - \tilde{f}(x)| < \frac{\epsilon}{\mu(\Omega)} \|f\|_{L^p(\mathbb{R}^n)}, \quad \forall x \in \Omega. \quad (2.6)$$

This completes the proof. $\square$

Remark 2.5 The space V_M is generated by $\{F_\gamma : \gamma \in \Gamma \cap M\}$ and the dimension d_M of V_M is at most the number of γ in M . Therefore, by the Assumption 1 and Lemma 2.4, we have

$$\begin{aligned} d_M \leq |\Gamma \cap M| &\leq \left(C(\Gamma)\mu(\Omega) + dN(\Gamma)N^n\right) \\ &\leq \mu(\Omega) \left(dN(\Gamma) \left(\frac{CN(\Gamma)B^{\frac{p'}{p}}}{\epsilon^{p'}}\right)^{\frac{1}{\alpha}} + C(\Gamma)\right) \\ &= C(\epsilon)\mu(\Omega), \end{aligned}$$

where $C(\epsilon) = \left(dN(\Gamma) \left(\frac{CN(\Gamma)B^{\frac{p'}{p}}}{\epsilon^{p'}}\right)^{\frac{1}{\alpha}} + C(\Gamma)\right).$

Remark 2.6 In the proof of the above lemma, if we do not choose M carefully, the bound of d_M may not be an effective estimation. For example, if d is the number of unit cubes that cover Ω , then $d > \mu(\Omega)$. On the other hand, if we consider a cube Q in $\mathbb{R}^n$ which contains Ω , then $\mu(Q) \geq \mu(\Omega)$. In both cases, we get a larger bound for d_M as $C\mu(\Omega)^2$ for some $C > 0$.

Remark 2.7 Lemma 2.4 implies that the set $V(\Omega, \delta) = \{f \in V^*(\Omega, \delta) : \|f\|_{L^p(\mathbb{R}^n)} = 1\}$ is totally bounded with respect to $\|\cdot\|_{L^\infty(\Omega)}$. Hence, following the similar argument in the related works [3, 32, 40, 41], the reader can verify that the random sampling inequality (1.2) holds for $V^*(\Omega, \delta)$ with positive probability. However, the effective number of samples in order to achieve the expected probability is of order $\mathcal{O}(\mu(\Omega)^2)$. The discretization technique used in [5, 11] constructs a class of piecewise constant functions, which “sandwiches” functions in a compact subset of V_M as well as estimates the bound of random variables accurately.

3 Discretization of functions

Let $(X, \|\cdot\|)$ be a Banach space. For a compact set K and a positive number ϵ , we define the covering number $N_\epsilon(A)$ as follows

$$N_\epsilon(K) := N_\epsilon(K, X) := \min \left\{ m : \exists f_1, f_2, \dots, f_m \in K, K \subseteq \bigcup_{j=1}^m B(f_j; \epsilon) \right\}.$$

The corresponding minimal ϵ -net is denoted by $\mathcal{N}_\epsilon(K, X)$, and $N_\epsilon(K, X) = |\mathcal{N}_\epsilon(K, X)|$. The following lemma is a well-known estimation of covering number of a closed ball in finite-dimensional space.

Lemma 3.1 ([10]) *Let X be a Banach space of dimension s . Then, the number of open balls of radius ω to cover $B(0; r)$ is bounded by $(\frac{2r}{\omega} + 1)^s$.*

Let M be a compact subset of $\mathbb{R}^n$. We consider the set

$$V_{M, \Omega} = \left\{ f \in V_M : \|f\|_{L^p(\mathbb{R}^n)}^p = \mu(\Omega) \right\}.$$

From (2.3), we see that $V_{M, \Omega}$ is a compact subset of $L^\infty(\mathbb{R}^n)$ and bounded by $D\mu(\Omega)^{\frac{1}{p}}$. Lemma 3.1 gives a bound of covering number of $V_{M, \Omega} \subseteq L^\infty(\mathbb{R}^n)$, i.e.,

$$N_\epsilon(V_{M, \Omega}, L^\infty(\mathbb{R}^n)) \leq \left(1 + \frac{2D\mu(\Omega)^{\frac{1}{p}}}{\epsilon} \right)^{d_M} \leq \left(\frac{4D\mu(\Omega)^{\frac{1}{p}}}{\epsilon} \right)^{d_M}. \quad (3.1)$$

In the following, we now discuss on discretization of functions in a compact set $V_{M, \Omega}$. Let $a \in (0, \frac{1}{2}]$ be a fixed small number (chosen later) and denote the set $\mathcal{A}_j = \mathcal{N}_{a(1+a)^j}(V_{M, \Omega}, L^\infty(\mathbb{R}^n))$ for $j \in \mathbb{Z}$. Let $j_0 \in \mathbb{Z}$ be a fixed integer (specified later), and for each $j(\geq j_0) \in \mathbb{Z}$, consider the map $A_j : V_{M, \Omega} \rightarrow \mathcal{A}_j$ such that $A_j(f)$ is a function in $\mathcal{A}_j$ closest to f with respect to $\|\cdot\|_{L^\infty(\mathbb{R}^n)}$. Then,

$$\|f - A_j(f)\|_{L^\infty(\mathbb{R}^n)} \leq a(1+a)^j. \quad (3.2)$$

If $f \in V_{M, \Omega}$ and $j \in \mathbb{Z} \cap (j_0, \infty)$, we construct a set of collection of points as follows.

$$U_j(f) := \left\{ x \in \mathbb{R}^n : |A_j(f)(x)| \geq (1+a)^j \right\}, \\ D_j(f) := U_j(f) \setminus \bigcup_{k>j} U_k(f), \quad D_{j_0}(f) := \mathbb{R}^n \setminus \bigcup_{k>j_0} U_k(f).$$

In particular, the set $D_j(f)$ is a collection of all points in $\mathbb{R}^n$ such that

$$(1+a)^j \leq |A_j(f)(x)|, \quad \text{and} \quad |A_k(f)(x)| < (1+a)^k, \quad \text{for all } k > j.$$

For each $f \in V_{M,\Omega}$, we define piecewise constant function $h(f)$ by

$$h(f) = \sum_{j>j_0} (1+a)^j \chi_{D_j(f)},$$

where χ_E is the characteristic function on E .

The following lemma gives some properties of the above h -mapping. We show that the absolute value of the function f in $V_{M,\Omega}$ is bounded between constant multiples of $h(f)$.

Lemma 3.2 For each $f \in V_{M,\Omega}$,

$$C_1(a)h(f)(x) \leq |f(x)| \leq C_2(a)h(f)(x) \quad \forall x \in \mathbb{R}^n \setminus D_{j_0}(f), \quad (3.3)$$

and

$$|f(x)| \leq C_2(a)(1+a)^{j_0} \quad \forall x \in D_{j_0}(f), \quad (3.4)$$

where $C_1(a) = (1-a)$ and $C_2(a) = (1+a)^2$.

Proof Let $x \in D_j(f)$ with $j > j_0$. Then, $x \in U_j(f)$ and $x \notin U_k(f)$ for $k > j$. Using (3.2) and from the definition of $U_j(f)$, we get

$$|f(x)| \geq |A_j(f)(x)| - a(1+a)^j \geq (1+a)^j - a(1+a)^j = C_1(a)(1+a)^j$$

and

$$|f(x)| \leq |A_{j+1}(f)(x)| + a(1+a)^{j+1} \leq (1+a)^{j+1} + a(1+a)^{j+1} = C_2(a)(1+a)^j,$$

where $C_1(a) = (1-a)$ and $C_2(a) = (1+a)^2$. Therefore, for all $x \in \mathbb{R}^n \setminus D_{j_0}(f) = \cup_{j>j_0} D_j(f)$ we have,

$$C_1(a)h(f)(x) \leq |f(x)| \leq C_2(a)h(f)(x).$$

The inequality (3.4) can be derived similarly. $\square$

Remark 3.3 For all $a \in (0, \frac{1}{2}]$, $C_1(a) \leq 1 \leq C_2(a)$, and $\lim_{a \rightarrow 0} C_1(a) = \lim_{a \rightarrow 0} C_2(a) = 1$. In the rest of the paper, we consider $a \in (0, \frac{1}{2}]$ with $\left(\frac{C_2(a)}{C_1(a)}\right)^p \leq \frac{5}{4}$.

Remark 3.4 For each $j > j_0$ and $f \in V_{M,\Omega}$, $A_j(f)$ is a bounded by $D\mu(\Omega)^{\frac{1}{p}}$. This implies, there exists $J \in \mathbb{Z}$ such that $C_1(a)(1+a)^J \leq D\mu(\Omega)^{\frac{1}{p}} < C_1(a)(1+a)^{J+1}$, and for each $j > J$, $D_j(f)$ is an empty set. This implies, $h(f)$ is the finite sum of piecewise constant functions. In fact, $\{h(f) : f \in V_{M,\Omega}\}$ is a finite collection of simple functions.

We list some notations which will be used in the rest of the following sections.

Symbol	Remark
$ X $ = The number of element in X	X is a finite set.
$N(\Gamma) = \sup_{k \in \mathbb{Z}^n} \Gamma \cap (k + [0, 1]^n) $	Intuitively, the maximum number of elements of Γ in a unit cube of $\mathbb{R}^n$.
$p' = \frac{p}{p-1}$	$p' = \infty$ for $p = 1$.
$C_1(a) = (1 - a)$	with a satisfying $\left(\frac{C_2(a)}{C_1(a)}\right)^p \leq \frac{5}{4}$.
$C_2(a) = (1 + a)^2$	

In the following lemma, we give a condition on h -mapping such that for functions in $V_{M,\Omega}$, the sample set discretize the integral norm on Ω . For a sample set $\xi = \{\xi_v\}_{v=1}^r$ and a function $f \in V$, denote

$$S(f, \xi) := (f(\xi_1), \dots, f(\xi_r)) \in \mathbb{R}^r, \quad \|S(f, \xi)\|_p^p := \frac{1}{r} \sum_{v=1}^r |f(\xi_v)|^p.$$

Lemma 3.5 *Let ξ be a sample set in Ω and $f \in V_{M,\Omega}$. Assume that the function $h(f)$ satisfies the following inequality*

$$\frac{1}{\mu(\Omega)} \|h(f)\|_{L^p(\Omega)}^p - \sigma \leq \|S(h(f), \xi)\|_p^p \leq \frac{1}{\mu(\Omega)} \|h(f)\|_{L^p(\Omega)}^p + \sigma, \quad (3.5)$$

for some $\sigma > 0$. Then,

$$\begin{aligned} C_1(a)^p \left(\frac{C_2(a)^{-p}}{\mu(\Omega)} \|f\|_{L^p(\Omega)}^p - (1+a)^{pj_0} - \sigma \right) \\ \leq \|S(f, \xi)\|_p^p \leq C_2(a)^p \left(\frac{C_1(a)^{-p}}{\mu(\Omega)} \|f\|_{L^p(\Omega)}^p + (1+a)^{pj_0} + \sigma \right). \end{aligned} \quad (3.6)$$

Proof For points on the set $D_{j_0}(f)$, the inequality (3.4) implies

$$\int_{D_{j_0}(f) \cap \Omega} |f(x)|^p dx \leq C_2(a)^p (1+a)^{pj_0} \mu(\Omega)$$

and

$$\frac{1}{r} \sum_{v: \xi_v \in D_{j_0}(f)} |f(\xi_v)|^p \leq C_2(a)^p (1+a)^{pj_0}.$$

By (3.3), we have

$$\begin{aligned}\|S(f, \xi)\|_p^p &\leq C_2(a)^P(1+a)^{pj_0} + C_2(a)^P\|S(h(f), \xi)\|_p^p \\ &\leq C_2(a)^P(1+a)^{pj_0} + \frac{C_2(a)^P}{\mu(\Omega)}\|h(f)\|_{L^p(\Omega)}^p + C_2(a)^P\sigma \\ &\leq C_2(a)^P\left(\frac{C_1(a)^{-P}}{\mu(\Omega)}\|f\|_{L^p(\Omega)}^p + (1+a)^{pj_0} + \sigma\right).\end{aligned}$$

On the other hand, we have

$$\begin{aligned}\|S(f, \xi)\|_p^p &\geq C_1(a)^P\|S(h(f), \xi)\|_p^p \\ &\geq C_1(a)^P\left(\frac{1}{\mu(\Omega)}\|h(f)\|_{L^p(\Omega)}^p - \sigma\right) \\ &\geq C_1(a)^P\left(\frac{C_2(a)^{-P}}{\mu(\Omega)}\int_{\Omega \setminus D_{j_0}(f)}|f(x)|^p dx - \sigma\right) \\ &\geq C_1(a)^P\left(\frac{C_2(a)^{-P}}{\mu(\Omega)}\|f\|_{L^p(\Omega)}^p - \frac{C_2(a)^{-P}}{\mu(\Omega)}\int_{D_{j_0}(f) \cap \Omega}|f(x)|^p dx - \sigma\right) \\ &\geq C_1(a)^P\left(\frac{C_2(a)^{-P}}{\mu(\Omega)}\|f\|_{L^p(\Omega)}^p - (1+a)^{pj_0} - \sigma\right).\end{aligned}$$

□

The lemma implies that the discretization of the integral norm of $f \in V_M$ and corresponding simple function $h(f)$ are related. We aim to find the probability for which the random sample set ξ satisfies the condition (3.5) on $h(f)$ for all $f \in V_{M,\Omega}$.

4 Random sampling

In this section, we discuss the main result of this paper. In order to derive the probabilistic estimates, we make use of the following lemma.

Lemma 4.1 ([6, Lemma 2.1]) *Let $\{g_v\}_{v=1}^r$ be independent random variables with zero mean on probability space (X, ρ) such that*

$$\|g_v\|_{L^1(X, \rho)} \leq 2, \quad \|g_v\|_{L^\infty(X, \rho)} \leq L, \quad 1 \leq v \leq r.$$

Then for any $\eta \in (0, 1)$, we have the following probability bound

$$\mathbb{P}\left\{\left|\sum_{v=1}^r g_v\right| \geq r\eta\right\} < 2\exp\left(-\frac{r\eta^2}{8L}\right).$$

It is easy to verify that the above lemma implies the following result.

Corollary 4.2 ([6, Corollary 2.2]) *Let $\xi = \{\xi_v\}_{v=1}^r$ be a random points drawn from probability space (X, ρ) and $\{\mathcal{F}_j\}_{j \in G}$ be a finite collection of finite set of functions in $L^1(X, \rho)$. Assume that for each $j \in G$ and $f \in \mathcal{F}_j$, we have*

$$\|f\|_{L^1(X, \rho)} \leq 1, \quad \|f\|_{L^\infty(X, \rho)} \leq L_j.$$

Then for each $j \in G$, for any $\eta_j \in (0, 1)$ and for all $f \in \mathcal{F}_j$, we have

$$\left| \|f\|_{L^1(X, \rho)} - \frac{1}{r} \sum_{v=1}^r |f(\xi_v)| \right| \leq \eta_j,$$

with probability at least $1 - 2 \sum_{j \in G} |\mathcal{F}_j| \exp\left(-\frac{r\eta_j^2}{8L_j}\right)$.

Let $p \in [1, \infty)$. For $j > j_0$, define

$$\mathcal{F}_j := \left\{ \frac{4}{5}(1+a)^{pj} \chi_{D_j(f)} : f \in V_{M, \Omega} \right\}.$$

By definition of $D_j(f)$, we consider those $j > j_0$ such that $C_1(a)(1+a)^j \leq D\mu(\Omega)^{\frac{1}{p}}$. Otherwise, $D_j(f)$ is empty. Choose $J \in \mathbb{Z}$ such that $C_1(a)(1+a)^J \leq D\mu(\Omega)^{\frac{1}{p}}$, i.e., $J \leq \frac{\log(D\mu(\Omega)^{\frac{1}{p}}/C_1(a))}{\log(1+a)}$.

Now, consider the index set $G = (j_0, J] \cap \mathbb{Z}$. Then, we apply Corollary 4.2 for the collection of sets $\{\mathcal{F}_j\}_{j \in G}$ and prove the following theorem.

Theorem 4.3 *Let $\xi = \{\xi_v\}_{v=1}^r$ be a sequence of i.i.d. random points that are uniformly distributed over the compact set Ω . If the sample size satisfies*

$$r \geq \frac{10}{\sigma^2} d_M |G|^2,$$

then for every $f \in V_M$ the following inequality

$$\begin{aligned} \frac{C_1(a)^p}{\mu(\Omega)} \left(C_2(a)^{-p} \|f\|_{L^p(\Omega)}^p - (1+a)^{pj_0} \|f\|_{L^p(\mathbb{R}^n)}^p - \sigma \|f\|_{L^p(\mathbb{R}^n)}^p \right) &\leq \|S(f, \xi)\|_p^p \\ &\leq \frac{C_2(a)^p}{\mu(\Omega)} \left(C_1(a)^{-p} \|f\|_{L^p(\Omega)}^p + (1+a)^{pj_0} \|f\|_{L^p(\mathbb{R}^n)}^p + \sigma \|f\|_{L^p(\mathbb{R}^n)}^p \right) \end{aligned}$$

holds with probability at least $1 - 2A_1|G| \exp\left(-\left(\frac{r\sigma^2}{|G|^2} - d_M\right)(1+a)^{-pj_0}\right)$, where $A_1 = \mathcal{O}(\exp(d_M \log \mu(\Omega)))$.

Proof Let $\xi = \{\xi_v\}_{v=1}^r$ be the random points uniformly, identically, and independently distributed over Ω with probability measure $d\rho = \frac{1}{\mu(\Omega)} dx$. Hence for each $j \in G$

and $\phi_j = \frac{4}{5}(1+a)^{pj} \chi_{D_j(f)}$, we have

$$\|\phi_j\|_{L^\infty(\Omega, \rho)} \leq \frac{4}{5}(1+a)^{pj} := L_j,$$

and

$$\begin{aligned} \|\phi_j\|_{L^1(\Omega, \rho)} &= \frac{4}{5\mu(\Omega)} \|(1+a)^{pj} \chi_{D_j(f)}\|_{L^1(\Omega)} \\ &= \frac{4C_1(a)^{-p}}{5\mu(\Omega)} \|f\|_{L^p(\Omega)}^p \quad (\text{by (3.3)}) \\ &\leq C_2(a)^{-p} \leq 1. \end{aligned}$$

Denote $\eta_j = \frac{4\sigma}{5|G|}$, then from Corollary 4.2, we have

$$\begin{aligned} \left| \frac{1}{r} \sum_{v=1}^r |\phi_j(\xi_v)| - \frac{1}{\mu(\Omega)} \|\phi_j\|_{L^1(\Omega)} \right| &\leq \eta_j \\ \frac{1}{\mu(\Omega)} \|\phi_j\|_{L^1(\Omega)} - \eta_j &\leq \frac{1}{r} \sum_{v=1}^r |\phi_j(\xi_v)| \leq \frac{1}{\mu(\Omega)} \|\phi_j\|_{L^1(\Omega)} + \eta_j, \end{aligned} \quad (4.1)$$

holds for each $j \in G$ and $\phi_j \in \mathcal{F}_j$ with minimum probability of $1 - 2 \sum_{j \in G} |\mathcal{F}_j| \exp\left(-\frac{r\eta_j^2}{8L_j}\right)$.

Since the sets $D_j(f)$ are pairwise disjoint, we get

$$\sum_{j \in G} \frac{1}{r} \sum_{v=1}^r |\phi_j(\xi_v)| = \frac{4}{5r} \sum_{v=1}^r \sum_{j \in G} (1+a)^{pj} \chi_{D_j(f)}(\xi_v) = \frac{4}{5} \|S(h(f), \xi)\|_p^p,$$

and

$$\sum_{j \in G} \|\phi_j\|_{L^1(C_R)} = \frac{4}{5} \int_{\Omega} \sum_{j \in G} (1+a)^{pj} \chi_{D_j(f)}(x) dx = \frac{4}{5} \|h(f)\|_{L^p(\Omega)}^p.$$

Hence, for all $f \in V_{M,\Omega}$, (4.1) implies

$$\begin{aligned} \frac{1}{\mu(\Omega)} \sum_{j \in G} \|\phi_j\|_{L^1(\Omega)} - \sum_{j \in G} \eta_j &\leq \sum_{j \in G} \frac{1}{r} \sum_{v=1}^r |\phi_j(\xi_v)| \leq \frac{1}{\mu(\Omega)} \sum_{j \in G} \|\phi_j\|_{L^1(\Omega)} + \sum_{j \in G} \eta_j \\ \frac{4}{5\mu(\Omega)} \|h(f)\|_{L^p(\Omega)}^p - \frac{4\sigma}{5} &\leq \frac{4}{5} \|S(h(f), \xi)\|_p^p \leq \frac{4}{5\mu(\Omega)} \|h(f)\|_{L^p(\Omega)}^p + \frac{4\sigma}{5} \\ \frac{1}{\mu(\Omega)} \|h(f)\|_{L^p(\Omega)}^p - \sigma &\leq \|S(h(f), \xi)\|_p^p \leq \frac{1}{\mu(\Omega)} \|h(f)\|_{L^p(\Omega)}^p + \sigma. \end{aligned}$$

Therefore, Lemma 3.5 implies that (3.6) hold for every $f \in V_{M,\Omega}$ with probability at least $1 - 2 \sum_{j \in G} |\mathcal{F}_j| \exp\left(-\frac{r\eta_j^2}{8L_j}\right)$.

In order to calculate the bound of $\sum_{j \in G} |\mathcal{F}_j| \exp\left(-\frac{r\eta_j^2}{8L_j}\right)$, we first evaluate the bound of $|\mathcal{F}_j|$. By the definition of $D_j(f)$ and construction of $\mathcal{F}_j$, we conclude that

$$|\mathcal{F}_j| \leq |\mathcal{A}_j|.$$

Therefore, from (3.1), we have

$$\begin{aligned} \log |\mathcal{F}_j| &\leq \log |\mathcal{A}_j| \leq d_M \log \left(\frac{4D\mu(\Omega)^{\frac{1}{p}}}{a(1+a)^j} \right) \\ &\leq d_M \log \left(\frac{4D\mu(\Omega)^{\frac{1}{p}}}{a} \right) + d_M(1+a)^{-pj} \\ &\leq \frac{d_M \log(C_4\mu(\Omega))}{p} + d_M(1+a)^{-pj}, \end{aligned}$$

where $C_4 = (\frac{4D}{a})^p$, and $A_1 = \exp(p^{-1}d_M \log(C_4\mu(\Omega)))$.

Now, taking summation over j , we get

$$\begin{aligned} \sum_{j \in G} |\mathcal{F}_j| \exp\left(-\frac{r\eta_j^2}{8L_j}\right) &\leq \sum_{j \in G} A_1 \exp\left(d_M(1+a)^{-pj} - \frac{r\eta_j^2}{8L_j}\right) \\ &= \sum_{j \in G} A_1 \exp\left(d_M(1+a)^{-pj} - \frac{r\sigma^2}{10|G|^2}(1+a)^{-pj}\right) \\ &= \sum_{j \in G} A_1 \exp\left(-R(1+a)^{-pj}\right) \\ &\leq A_1|G| \exp\left(-R(1+a)^{-pj_0}\right), \end{aligned}$$

where $R = \frac{r\sigma^2}{10|G|^2} - d_M > 0$ for sufficiently large sample size r .

Let $f \in V_M \setminus \{0\}$ be arbitrary. Then, $g = \frac{f\mu(\Omega)^{\frac{1}{p}}}{\|f\|_{L^p(\mathbb{R}^n)}} \in V_{M,\Omega}$ and satisfies (3.6). Therefore,

$$\begin{aligned} \frac{C_1(a)^p}{\mu(\Omega)} \left(C_2(a)^{-p} \|f\|_{L^p(\Omega)}^p - (1+a)^{pj_0} \|f\|_{L^p(\mathbb{R}^n)}^p - \sigma \|f\|_{L^p(\mathbb{R}^n)}^p \right) &\leq \|S(f, \xi)\|_p^p \\ &\leq \frac{C_2(a)^p}{\mu(\Omega)} \left(C_1(a)^{-p} \|f\|_{L^p(\Omega)}^p + (1+a)^{pj_0} \|f\|_{L^p(\mathbb{R}^n)}^p + \sigma \|f\|_{L^p(\mathbb{R}^n)}^p \right), \end{aligned}$$

holds with probability at least $1 - 2A_1|G|\exp\left(-R(1+a)^{-pj_0}\right)$ with an additional assumption $R > 0$. $\square$

Theorem 4.4 Let $\xi = \{\xi_v : v = 1, 2, \dots, r\}$ be a sequence of independent random variables that are drawn uniformly from Ω . Suppose that $0 < \tau < 1$ is small enough and if the number of sample size r satisfies

$$r \geq \frac{10}{\sigma^2} d_M |G|^2 = \mathcal{O}(\mu(\Omega)(\log \mu(\Omega))^2),$$

then for every $f \in V^*(\Omega, \delta)$ the sampling inequality

$$\frac{r}{\mu(\Omega)} \left(\frac{1-\delta}{5} - \frac{p2^{p+1}\tau}{5} - pD^{p-1}\tau \right) \|f\|_{L^p(\mathbb{R}^n)}^p \leq \sum_{v=1}^r |f(\xi_v)|^p \leq \frac{r}{\mu(\Omega)} \left(\frac{2B}{A} + pD^{p-1} \right) \|f\|_{L^p(\mathbb{R}^n)}^p, \quad (4.2)$$

holds with probability at least $1 - 2A_1|G|\exp\left(-R(1+a)^{-pj_0}\right)$.

Proof Without loss of generality, let $f \in V^*(\Omega, \delta)$ with $\|f\|_{L^p(\mathbb{R}^n)} = 1$.

From Lemma 2.4, for $\tau > 0$, there exist compact set M and $\tilde{f} \in V_M$ such that

$$\begin{aligned} \|f - \tilde{f}\|_{L^\infty(\Omega)} &< \frac{\tau}{\mu(\Omega)}, \\ \|f - \tilde{f}\|_{L^p(\Omega)} &< \tau, \end{aligned}$$

and for $x \in \mathbb{R}^n$,

$$\begin{aligned} \tilde{f}(x) &= \sum_{\gamma \in \Gamma \cap M} c_\gamma F_\gamma(x), \\ |\tilde{f}(x)| &\leq \left(\sum_{\gamma \in \Gamma \cap M} |c_\gamma|^p \right)^{\frac{1}{p}} \left(\sum_{\gamma \in \Gamma \cap M} |F_\gamma(x)|^{p'} \right)^{\frac{1}{p'}} \\ &\leq B^{\frac{1}{p}} \|f\|_{L^p(\mathbb{R}^n)} \left(\sum_{\gamma \in \Gamma} |\Theta(x - \gamma)|^{p'} \right)^{\frac{1}{p'}} \\ &\leq D \|f\|_{L^p(\mathbb{R}^n)}. \end{aligned}$$

For given positive numbers a and b , $|a^p - b^p| \leq p(\max\{a, b\})^{p-1}|a - b|$. This implies

$$\begin{aligned} \left| \|f\|_{L^p(\Omega)}^p - \|\tilde{f}\|_{L^p(\Omega)}^p \right| &\leq p(1 + \tau)^{p-1}\tau \leq p2^{p-1}\tau \\ \|f\|_{L^p(\Omega)}^p - p2^{p-1}\tau &\leq \|\tilde{f}\|_{L^p(\Omega)}^p \leq \|f\|_{L^p(\Omega)}^p + p2^{p-1}\tau, \end{aligned} \quad (4.3)$$

and

$$\begin{aligned} \left| |f(\xi_v)|^p - |\tilde{f}(\xi_v)|^p \right| &\leq p \left(\max\{|f(\xi_v)|, |\tilde{f}(\xi_v)|\} \right)^{p-1} |f(\xi_v) - \tilde{f}(\xi_v)| \\ &\leq \frac{pD^{p-1}\tau}{\mu(\Omega)}. \end{aligned}$$

Therefore,

$$\sum_{v=1}^r |\tilde{f}(\xi_v)|^p - \frac{rpD^{p-1}\tau}{\mu(\Omega)} \leq \sum_{v=1}^r |f(\xi_v)|^p \leq \sum_{v=1}^r |\tilde{f}(\xi_v)|^p + \frac{rpD^{p-1}\tau}{\mu(\Omega)}. \quad (4.4)$$

Since $V_M \subseteq V$ and using (2.2), we get

$$\|\tilde{f}\|_{L^p(\mathbb{R}^n)}^p \leq \frac{1}{A} \sum_{\gamma \in \Gamma \cap M} |c_\gamma|^p \leq \frac{1}{A} \sum_{\gamma \in \Gamma} |c_\gamma|^p \leq \frac{B}{A} \|f\|_{L^p(\mathbb{R}^n)}^p.$$

Hence, the Theorem 4.3 and the inequality (4.3) and (4.4) follow that

$$\begin{aligned} \frac{r}{\mu(\Omega)} C_1(a)^p \left(C_2(a)^{-p} \|\tilde{f}\|_{L^p(\Omega)}^p - (1+a)^{pj_0} \|\tilde{f}\|_{L^p(\mathbb{R}^n)}^p - \sigma \|\tilde{f}\|_{L^p(\mathbb{R}^n)}^p \right) &\leq \sum_{v=1}^r |\tilde{f}(\xi_v)|^p \\ &\leq \frac{r}{\mu(\Omega)} C_2(a)^p \left(C_1(a)^{-p} \|\tilde{f}\|_{L^p(\Omega)}^p + (1+a)^{pj_0} \|\tilde{f}\|_{L^p(\mathbb{R}^n)}^p + \sigma \|\tilde{f}\|_{L^p(\mathbb{R}^n)}^p \right) \\ \frac{r}{\mu(\Omega)} C_1(a)^p \left(C_2(a)^{-p} \|f\|_{L^p(\Omega)}^p - C_2(a)^{-p} p2^{p-1}\tau - (1+a)^{pj_0} \frac{B}{A} - \sigma \frac{B}{A} \right) \\ &\leq \sum_{v=1}^r |\tilde{f}(\xi_v)|^p \leq \frac{r}{\mu(\Omega)} C_2(a)^p \left(C_1(a)^{-p} + (1+a)^{pj_0} + \sigma \right) \frac{B}{A} \\ \frac{r}{\mu(\Omega)} \left[C_1(a)^p \left(C_2(a)^{-p} (1-\delta) - (1+a)^{pj_0} \frac{B}{A} - \sigma \frac{B}{A} \right) - \frac{4}{5} p2^{p-1}\tau - pD^{p-1}\tau \right] \\ &\leq \sum_{v=1}^r |f(\xi_v)|^p \leq \frac{r}{\mu(\Omega)} \left[C_2(a)^p \left(C_1(a)^{-p} + (1+a)^{pj_0} + \sigma \right) \frac{B}{A} + pD^{p-1}\tau \right]. \end{aligned} \quad (4.5)$$

Now, we choose j_0 and σ such that

$$\frac{\sigma B}{A} = \frac{C_2(a)^{-p}(1-\delta)}{2},$$

$$\frac{B(1+a)^{pj_0}}{A} = \frac{C_2(a)^{-p}(1-\delta)}{4}.$$

This implies,

$$|j_0| = \log \left(\frac{4BC_2(a)^p}{A(1-\delta)} \right) / p \log(1+a).$$

and

$$\begin{aligned} |G| &\leq J + |j_0| \leq \log(D^p C_1(a)^{-p} \mu(\Omega)) / p \log(1+a) \\ &\quad + \log \left(\frac{4BC_2(a)^p}{A(1-\delta)} \right) / p \log(1+a) \\ &\leq \frac{\log(C_5 \mu(\Omega))}{p \log(1+a)}, \end{aligned}$$

where $C_5 = \frac{5BD^p}{A(1-\delta)}$. Also, $\frac{C_2(a)^p}{C_1(a)^p} \leq \frac{5}{4}$.

Hence, bounds of the sampling inequality (4.5) can be revised. We get the lower estimate as

$$\begin{aligned} &\frac{r}{\mu(\Omega)} \left(\frac{1-\delta}{5} - \frac{4}{5} p 2^{p-1} \tau - p D^{p-1} \tau \right) \\ &\leq \frac{r}{\mu(\Omega)} \left(\frac{4C_1(a)^p}{5C_2(a)^p} (1-\delta) - \frac{4}{5} p 2^{p-1} \tau - p D^{p-1} \tau \right), \end{aligned}$$

and upper bound as

$$\frac{r}{\mu(\Omega)} \left(\frac{5B}{4A} + \frac{1-\delta}{4} + \frac{1-\delta}{2} + p D^{p-1} \right) \leq \frac{r}{\mu(\Omega)} \left(\frac{2B}{A} + p D^{p-1} \right).$$

This implies for every $f \in V^*(\Omega, \delta)$

$$\frac{r}{\mu(\Omega)} \left(\frac{1-\delta}{5} - \frac{p 2^{p+1} \tau}{5} - p D^{p-1} \tau \right) \|f\|_{L^p(\mathbb{R}^n)}^p \leq \sum_{v=1}^r |f(\xi_v)|^p \leq \frac{r}{\mu(\Omega)} \left(\frac{2B}{A} + p D^{p-1} \right) \|f\|_{L^p(\mathbb{R}^n)}^p,$$

holds with probability at least $1 - 2A_1 |G| \exp \left(-R(1+a)^{-pj_0} \right)$. $\square$

Remark 4.5 Let $\{\xi_v\}$ be random sample drawn uniformly from Ω . The sampling inequality (4.2) holds with probability at least $1 - \epsilon$ if

$$\begin{aligned} 2A_1|G|\exp\left(-R(1+a)^{-pj_0}\right) \\ \leq 2|G|\exp\left(d_M p^{-1} \log(C_4 \mu(\Omega)) - (1+a)^{-pj_0} \left(\frac{r\sigma^2}{10|G|^2} - d_M\right)\right) \\ \leq 2 \frac{\log(C_5 \mu(\Omega))}{p \log(1+a)} \exp\left(C(\tau) \mu(\Omega) p^{-1} \log(C_4 \mu(\Omega)) \right. \\ \left. - (1+a)^{-pj_0} \left(\frac{r\sigma^2 p^2 (\log(1+a))^2}{(\log(C_5 \mu(\Omega)))^2} - C(\tau) \mu(\Omega)\right)\right) \\ < \epsilon. \end{aligned}$$

Hence, if the sample size $r \geq \mathcal{O}(\mu(\Omega)(\log \mu(\Omega))^3)$, then sampling inequality (4.2) holds with probability at least $1 - \epsilon$.

Function reconstruction from their random samples was studied in [28, 29, 32, 40]. The reconstruction method for the localized reproducing kernel space follows in a similar line of proof in [32, 40]. For the sake of exposition, we provide the reconstruction algorithm in the following.

As discussed in Sect. 2, the finite-dimensional subspace V_M is an approximation of V over Ω . Thus, for computational simplicity, we discuss the reconstruction of functions in V_M from $(\xi_v, f(\xi_v))_{v=1}^r$.

Theorem 4.6 Let $\xi = \{\xi_v : v = 1, 2, \dots, r\}$ be a sequence of independent random variable that are drawn uniformly from Ω . For a compact subset M of $\mathbb{R}^n$, suppose $\{F_\gamma : \gamma \in \Gamma \cap M\}$ is linearly independent over Ω , i.e., there exists $w > 0$ such that

$$w \sum_{\gamma \in \Gamma \cap M} |c_\gamma|^p \leq \left\| \sum_{\gamma \in \Gamma \cap M} c_\gamma F_\gamma \right\|_{L^p(\Omega)}^p. \quad (4.6)$$

Then for $\tilde{\sigma} \in (0, \frac{Aw}{2})$,

$$\frac{4}{5\mu(\Omega)} \left(\frac{Aw}{2} - \tilde{\sigma} \right) \|f\|_{L^p(\mathbb{R}^n)}^p \leq \|S(f, \xi)\|_p^p \leq \frac{1}{\mu(\Omega)} \left(\frac{Aw}{2} + \frac{5}{4} + \tilde{\sigma} \right) \|f\|_{L^p(\mathbb{R}^n)}^p, \quad (4.7)$$

for all $f \in V_M$ with probability at least $1 - 2A_1|G|\exp\left(- (r|G|^{-2}C_2(a)^{2p} - d_M) \frac{2C_2(a)^p}{Aw}\right)$, where A_1 is the same as in Theorem 4.3 and $|G| \leq \frac{\log(5D^p\mu(\Omega)) - \log(8Aw)}{p \log(1+a)}$.

Moreover, there exists a set of reconstruction functions $(S_v)_{v=1}^r$ such that for all $f \in V_M$,

$$f(x) = \sum_{v=1}^r f(\xi_v) S_v(x), \quad x \in \Omega.$$

Proof Theorem 4.3 and the Riesz basis condition (2.2) imply that with probability at least $1 - 2A_1|G|\exp\left(-\left(\frac{r\sigma^2}{|G|^2} - d_M\right)(1+a)^{-pj_0}\right)$,

$$\begin{aligned} \frac{C_1(a)^p}{\mu(\Omega)} \left(C_2(a)^{-p} Aw - (1+a)^{pj_0} - \sigma \right) \|f\|_{L^p(\mathbb{R}^n)}^p &\leq \|S(f, \xi)\|_p^p \\ &\leq \frac{C_2(a)^p}{\mu(\Omega)} \left(C_1(a)^{-p} + (1+a)^{pj_0} + \sigma \right) \|f\|_{L^p(\mathbb{R}^n)}^p \end{aligned}$$

holds for all $f \in V_M$. For the choice,

$$(1+a)^{pj_0} = \frac{Aw}{2C_2(a)^p}, \quad \text{and} \quad \tilde{\sigma} = \frac{\sigma}{C_2(a)^p}.$$

Then for all f in V_M , the sampling inequality (4.7) holds with probability at least $1 - 2A_1|G|\exp\left(-\left(\frac{r\tilde{\sigma}^2 C_2(a)^{2p}}{|G|^2} - d_M\right)\frac{2C_2(a)^p}{Aw}\right)$, where $|G| \leq \frac{\log(5D^p\mu(\Omega)) - \log(8Aw)}{p \log(1+a)}$.

For $f = \sum_{\gamma \in \Gamma \cap M} c_\gamma F_\gamma$ be an arbitrary function in V_M and given random data $(\xi_v, f(\xi_v))_{v=1}^r$, we have the following system of linear equation.

$$f(\xi_v) = \sum_{\gamma \in \Gamma \cap M} c_\gamma F_\gamma(\xi_v), \quad v = 1, 2, \dots, r.$$

The linear system can be rewritten as $Lc = d$, where $(L)_{v,\gamma} = F_\gamma(\xi_v)$ and $d = (f(\xi_v))_{1 \leq v \leq r}^T$ is a column matrix. By (2.2) and (4.7), we deduce that with the same probability,

$$\frac{4r}{5B\mu(\Omega)} \left(\frac{Aw}{2} - \tilde{\sigma} \right) \|c\|_{\ell^p(\Gamma \cap M)}^p \leq \sum_{v=1}^r |f(\xi_v)|^p = \|Lc\|_{\ell^p(\Gamma \cap M)}^p, \quad \forall c \in \ell^p(\Gamma \cap M).$$

Therefore, $L^T L$ is invertible and we have the following reconstruction formula

$$f(x) = \sum_{v=1}^r f(\xi_v) S_v(x), \quad x \in \Omega,$$

where $(S_v(x))_{1 \leq v \leq r}^T = L(L^T L)^{-1} (F_\gamma(x))_{\gamma \in \Gamma \cap M}^T$. □

Acknowledgements We are grateful to the anonymous referee for meticulously reading the manuscript and giving us helpful suggestions that helped to improve the quality of the paper.

Funding The first author was financially supported by the Council of Scientific & Industrial Research. The second author acknowledges the financial support of the Science and Engineering Research Board, Government of India, through project no. CRG/2019/002412.

Declarations

Conflict of interest The authors declare no competing interests.

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